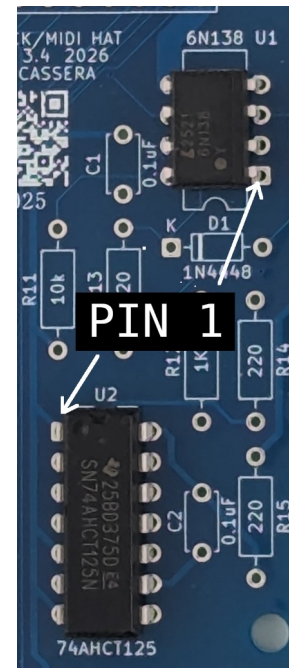


Assembling the Joystick/MIDI HAT

All components except for the 32 pin header socket (J1) will be mounted to the front of the board. The front is defined as the side with the QR code on it.

For the most part, the components can be installed in whatever order you would like. These instructions are the order in which I assemble the boards. All component positions and values are marked on the board.

1. Solder resistors R1 through R11 to the board. These are the pack of 11, 10k Ω resistors from the kit. On the board they are the line of 10 resistors along the center-left side of the board, plus 1 in the center of the board.
2. Solder R13 through R15 to the board. These are the 220 Ω and 1K Ω resistors. R12 is the 1K Ω resistor while R13 through R15 are the 220 Ω resistors.
3. Solder D1 to the board. This is the single diode in the kit. **The diode needs to be installed facing a certain direction. Make sure the black band on the diode matches the designated line on the PCB.**
4. Solder chip U1 to the board. **Be sure to position the chip correctly. Pin 1 is located on the side of the chip with the long recessed edge.**
5. Solder chip U2 to the board. **Be sure to position the chip correctly. Pin 1 is located on the side of the chip with the notch removed from the short edge.**
6. Install C1 & C2 capacitors.
7. Install pin headers at J6 through J8.
8. Install J1 header socket. **This is the 32 pin socket that needs to be installed on the opposite side of the board from all the other components. This socket needs to be as flat as possible to the board for alignment purposes when mounting the HAT to your JR2.**
9. Install the two DE-9 connectors at J2 and J3. **You may need to bend the large prongs a little to make it all line up properly. These prongs do not need to be soldered.**
10. Install the two 5 pin din connectors at J4 and J5. **You may need to cut or break off the two prongs towards the front of the connectors as they may not line up with the hole on the PCB. These are not needed for any electrical connection.**



Joystick/MIDI HAT

Parts list:

1)	HAT PCB	1	
2)	32 pin header socket	1	(J1)
3)	6 pin header	2	(J7, J8)
4)	4 pin header	1	(J6)
5)	DE-9 connectors	2	(J2, J3)
6)	5 pin din connectors	2	(J4, J5)
7)	10K Ω resistors	11	(R1-R11)
8)	1K Ω resistor	1	(R12)
9)	220 Ω resistors	3	(R13-R15)
10)	.1 uF capacitors	2	(C1, C2)
11)	1N4448 diode	1	(D1)
12)	6N138 optocoupler (8 pin chip)	1	(U1)
13)	74AHCT125 non-inverting buffer (14 pin chip)	1	(U2)

